laboratory scale may work poorly or may not work at all on industrial scale. Therefore it is not always possible to blindly apply the laboratory conditions of a fermentation technique developed to industry.

At the laboratory scale, one is interested in the maximum yield of the product for unit time. At the industry level, besides the product yield, minimal operating cost is another important factor for consideration.

Preparation and sterilization of media

According to the specific industrial production basic components needed to carry out fermentation are selected as per the required volume.

Medium components should be free from contamination. So all the medium components employed in the fermentation process are sterilized. Sterilization is mostly carried out by applying heat and to lesser extent other physical methods, chemical methods (disinfectants) and radiation (using UV rays, γ rays). Batch Sterilization is carried out at 121°C (20 to 60 mins) where as continuous sterilization is done at 140°C for (30 to 120 secs). Much energy is wasted on batch sterilization on compared with continuous sterilization nearly 80 to 90% of energy saved during this process. Air and heat sensitive components are sterilizied by membrane filters.

Fermentation Process

It involves the propagation of the microorganism and the production of the desired product. Fermentation process is divided depending on the feeding strategy of the culture and medium as follows.

i. Batch Fermentation
ii. Continuous Fermentation
iii. Fed batch Fermentation

i. Batch Fermentation

The medium and culture are initially fed into the vessel and it is then closed. After that, no components are added apart from Oxygen. The pH is adjusted during the course of process by adding either acid or alkali. The fermentation is allowed to run for a predetermined period of time and the product is harvested at the end. Foaming is controlled by adding antifoam agents such as palm oil or soybeans oil. Heat generated is regulated by providing water circulation system around the vessel for heat exchange.

ii. Continuous fermentation

This is an open system. It involves the removal of culture medium continuously and replacement of them with a fresh sterile medium in a bioreactor.

In this method, homogenously mixing reactors which include chemo stat and turbid stat bioreactors are used. Examples: production of antibiotics, organic solvents, beer, ethanol and SCP.

iii. Fed batch system

It is a combination of both batch and continuous systems. In this, additional nutrients are added to the fermentors as the fermentation is in progress. This extends the time of operation, but the products are harvested at the end of the production cycle as in batch fermenter.

HOTS

Why does industry prefer continous culture?

Followed by the fermentation, production, products are harvested or separated by downstream processing.

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